



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Developing AI Apps and Agents on Azure (AI-103)

TGS Ref No: TGS-2023036651

Conducted by

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Version 4.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
2.0	26 Aug 2026	Prior AI-102 courseware release	Tertiary Infotech Academy Pte Ltd
3.0	27 Sep 2026	AI-103 title transition	Tertiary Infotech Academy Pte Ltd
4.0	27 Sep 2026	Updated to v4.0 deck slide map, AI-103 scope, 10 mechanism-led labs, and approved 60+60 minute assessments.	Tertiary Infotech Academy Pte Ltd

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Course Overview

A two-day instructor-led WSQ course that develops the ability to analyse, evaluate, and improve Azure AI solutions through Microsoft Foundry services, governed engineering decisions, and observable evidence.

Learning Outcomes

- LO1: Analyse algorithms in Microsoft Azure AI applications and establish their correlation with efficiency.
- LO2: Identify and evaluate strengths and limitations of Microsoft Azure AI applications.
- LO3: Assess feasibility and improvements when applying Azure AI to product and maintenance processes.

Daily Schedule

Time	Duration	Topic / Activity	Method	Slides
Day 1 · 09:00–09:40	40 min	Administration, outcomes, assessment briefing	Facilitated briefing	Slides 1–15
Day 1 · 09:40–10:45	55–70 min	Topic/Lab 01: Plan, Manage, Monitor, and Secure an Azure AI Solution	Mechanism + evidence lab	Slides 17–26
Day 1 · 11:00–12:05	55–70 min	Topic/Lab 02: Responsible AI, Content Safety, Governance	Mechanism + evidence lab	Slides 27–37
Day 1 · 12:05–13:00	55–70 min	Topic/Lab 03: Generative AI, Foundry, Azure OpenAI, Foundry Workflow, RAG	Mechanism + evidence lab	Slides 38–47
Day 1 · 14:00–15:05	55–70 min	Topic/Lab 04: Agentic AI, Foundry Agent Service, Multi-Agent Concepts	Mechanism + evidence lab	Slides 48–57
Day 1 · 15:05–16:10	55–70 min	Topic/Lab 05: Image and Video Generation, Multimodal Understanding	Mechanism + evidence lab	Slides 58–67
Day 1 · 16:25–17:35	55–70 min	Topic/Lab 06: NLP, Language, Speech, Translation	Mechanism + evidence lab	Slides 68–77
Day 1 · 10:45–11:00	15 min	Morning break	Break	—
Day 1 · 13:00–14:00	60 min	Lunch	Break	—
Day 1 · 16:10–16:25	15 min	Afternoon break	Break	—
Day 1 · 17:35–18:00	25 min	Evidence review and recap	Discussion	Slides 17–77
Day 2 · 09:00–09:15	15 min	Digital attendance and recap	Review	Slide 78
Day 2 · 09:15–10:20	65–75 min	Topic/Lab 07: Generative Text Analysis and Translation	Mechanism + evidence lab	Slides 79–88
Day 2 · 10:35–11:45	65–75 min	Topic/Lab 08: Azure AI Search, Knowledge Mining, Vector Search	Mechanism + evidence lab	Slides 89–98
Day 2 · 11:45–13:00	65–75 min	Topic/Lab 09: Document Intelligence and Content Understanding	Mechanism + evidence lab	Slides 99–109
Day 2 · 14:00–15:15	65–75 min	Topic/Lab 10: AI-103 Capstone and Course Review	Mechanism + evidence lab	Slides 110–120
Day 2 · 10:20–10:35	15 min	Morning break	Break	—
Day 2 · 13:00–14:00	60 min	Lunch	Break	—
Day 2 · 15:15–15:45	30 min	Assessment briefing and preparation	Briefing	Slides 11–13
Day 2 · 16:00–18:00	120 min	WA-SAQ (60 min) + PP (60 min)	Open-book assessment	Slides 123–125

Topic-by-Topic Breakdown

Topic 01 - Plan, secure, and operate Foundry services • Slides 17–26

Lab 01 - Plan, Manage, Monitor, and Secure an Azure AI Solution • Slide 25

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 02 - Responsible AI, content safety, and governance • Slides 27–37

Lab 02 - Responsible AI, Content Safety, Governance • Slide 36

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 03 - Generative AI apps and grounded RAG • Slides 38–47

Lab 03 - Generative AI, Foundry, Azure OpenAI, Foundry Workflow, RAG • Slide 46

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 04 - Agents, tools, memory, and multi-agent control • Slides 48–57

Lab 04 - Agentic AI, Foundry Agent Service, Multi-Agent Concepts • Slide 56

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 05 - Image/video generation and multimodal understanding • Slides 58–67

Lab 05 - Image and Video Generation, Multimodal Understanding • Slide 66

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 06 - Language, speech, translation, and SSML • Slides 68–77

Lab 06 - NLP, Language, Speech, Translation • Slide 76

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 07 - Generative text analysis and translation • Slides 79–88

Lab 07 - Generative Text Analysis and Translation • Slide 87

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 08 - Knowledge retrieval, indexing, and grounding • Slides 89–98

Lab 08 - Azure AI Search, Knowledge Mining, Vector Search • Slide 97

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 09 - Document Intelligence and Content Understanding • Slides 99–109

Lab 09 - Document Intelligence and Content Understanding • Slide 108

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Topic 10 - Capstone: production AI solution decision record • Slides 110–120

Lab 10 - AI-103 Capstone and Course Review • Slide 119

Coverage: architecture and boundaries; runtime mechanism; exact fields/endpoints; implementation excerpt; editable metric; decision rule; failure/control; activity bridge; acceptance evidence.

Resources Required

- Trainer PPTX and learner PDF
- Learner Guide
- Azure subscription or trainer-provided lab environment
- Modern browser, Azure CLI, Python 3, and Visual Studio Code as required
- LMS/TMS access for courseware and assessment

Assessment

Approved instruments: Written Assessment - Short-Answer Questions (WA-SAQ), 60 minutes, K1–K6; Practical Performance (PP), 60 minutes, A1–A6. Both are individual and open book. Complete assessment digital attendance before starting and submit candidate papers on the LMS.